Embedded C Lab DA 5

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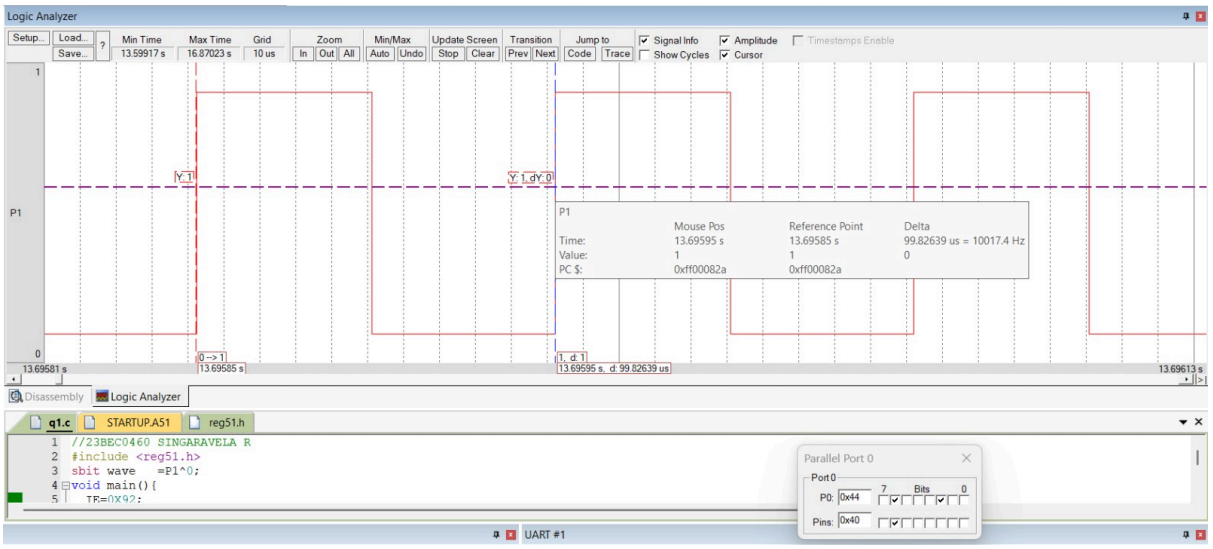
23BEC0460

Write a 8051 based Embedded C program using interrupts to do the following: a) Receive data serially and sent it to P0, b) Make timer 0 generate a square wave of 100micro second time period on P1.0. Assume that XTAL-11.0592. Set the baud rate at 4800.

q1.c | STARTUP.A51 | reg51.h | q2.c

```c
//23BEC0460 SINGARAVELA R
#include <reg51.h>
sbit wave   =P1^0;
void main(){
  IE=0X92;
  TMOD=0x22;
  TH0=-46;
  TH1=-6;
  TR1=1;
  SCON=0X50;
  TR0=1;
  while(1){

  }
}
void serial() interrupt 4{
  if (RI == 1) {
        P0 = SBUF;
        RI = 0;
    }
}
void timer0() interrupt 1{
    wave=~wave;
}
```

Output:



Code :

```
//23BEC0460 SINGARAVELA R
#include <reg51.h>
sbit wave      =P1^0;
void main(){
        IE=0X92;
        TMOD=0x22;
        TH0=-46;
        TH1=-6;
        TR1=1;
        SCON=0X50;
  TR0=1;
        while(1){

        }
}
void serial() interrupt 4{
        if (RI == 1) {
      P0 = SBUF;
      RI = 0;
   }
}
void timer0() interrupt 1{
   wave=~wave;
}
```

Calculation:

Task 5 A:

Square wave delay calculation:

$$\frac{100 \times 10^{-6}}{1.0852 \times 10^{-6} \times 2} = 46$$

$65535 - 46 = 65489$

$= FFD1$

THO = FF TLO = 48

THO = -46

Serial communication: TH1 = -6

SCON:

D	1	0	1	0	0	0	0

= 0x50

Mode 1 receive enable

IE:

1	0	0	1	0	0	0	0

9 0

- 0x92

TMOD:

0	0	1	0	0	0	0	0

2 1

- 0x22

2.Write a 8051 based Embedded C program using timer 0 to generate a 500 Hz and timer 1 to 7kHz square wave frequency on P1.0 and P1.1 respectively using Interrupts. Then examine the frequency using the KEIL IDE inbuilt Logic Analyzer.

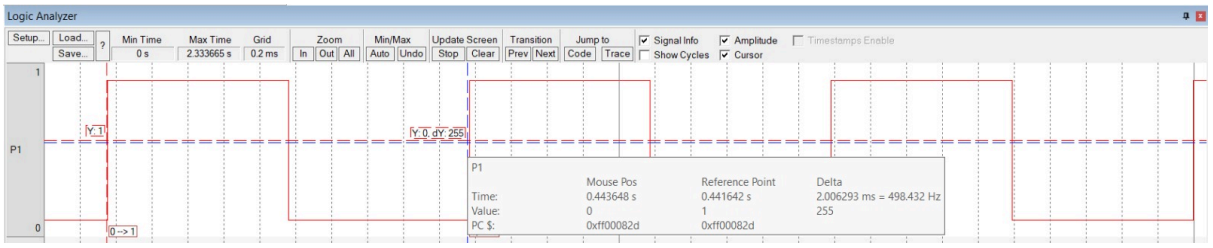
Code:

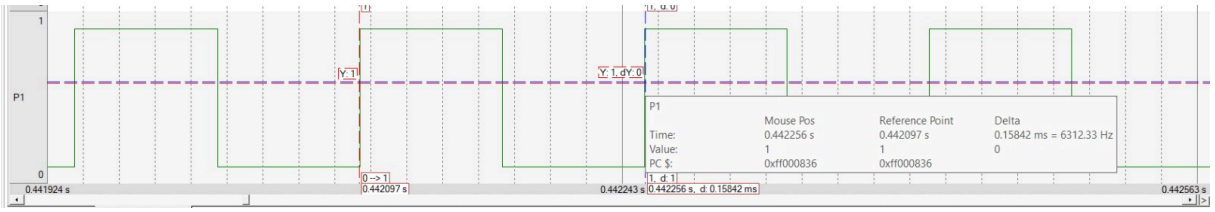
```
//23BEC0460 SINGARAVELA R
#include <reg51.h>
sbit P1_0 = P1^0;
sbit P1_1 = P1^1;

void timer0_ISR(void) interrupt 1 {
    TH0 = 0xFC;
    TL0 = 0x66;
    P1_0 = ~P1_0;
}

void timer1_ISR(void) interrupt 3 {
    TH1 = 0xFF;
    TL1 = 0xBE;
    P1_1 = ~P1_1;
}

void main(void) {
    TMOD = 0x11;

    TH0 = 0xFC;
    TL0 = 0x66;

    TH1 = 0xFF;
    TL1 = 0xBE;

    ET0 = 1;
    ET1 = 1;
    EA = 1;

    TR0 = 1;
    TR1 = 1;

    while(1);
}
```

q1.c | STARTUP.A51 | reg51.h | q2.c

```c
//23BEC0460 SINGARAVELA R
#include <reg51.h>
sbit P1_0 = P1^0;
sbit P1_1 = P1^1;

void timer0_ISR(void) interrupt 1 {
    TH0 = 0xFC;
    TL0 = 0x66;
    P1_0 = ~P1_0;
}

void timer1_ISR(void) interrupt 3 {
    TH1 = 0xFF;
    TL1 = 0xBE;
    P1_1 = ~P1_1;
}

void main(void) {
    TMOD = 0x11;

    TH0 = 0xFC;
    TL0 = 0x66;

    TH1 = 0xFF;
    TL1 = 0xBE;

    ET0 = 1;
    ET1 = 1;
    EA = 1;

    TR0 = 1;
    TR1 = 1;

    while(1);
}
```

Output:





Calculation:

Task 5B:

Square wave calculation:

500 Hz = $\frac{1}{2 \times 500 \times 1.085 \times 10^{-6}}$ = 921 => 65535 - 921 = 64614 = FC66

7 kHz = $\frac{1}{2 \times 7 \times 10^{3} \times 1.085 \times 10^{-6}}$ = 66 . 65535 - 66 = 65469 = FFBD

IE:

1	0	0	0	1	0	1	0

- 0x8A

TMOD:

0	0	0	1	0	0	0	1

- 0x11

TH0 = FC TL0 = 66

TH1 = FF TH0 = BD